

Fleet deployment and freight allocation for cleaner maritime logistics under demand uncertainty[☆]

Xiuwen Wang, Lu Zhen^{*}

School of Management, Shanghai University, Shanghai 200444, China

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ABSTRACT

Maritime shipping accounts for the majority of global trade but remains one of the largest contributors to greenhouse gas emissions. In response to increasingly stringent environmental regulations, ship operators are adopting various green technologies, including scrubber-equipped vessels, dual-fuel retrofitted vessels, and biofuel single-fuel vessels, each with distinct implications for cost, capacity, and environmental performance. At the same time, shipping demand is subject to considerable uncertainty due to volatile trade flows, geopolitical shocks, and supply chain disruptions. This paper addresses the problem of fleet deployment and freight allocation for cleaner maritime logistics under demand uncertainty. We develop a two-stage stochastic programming model, where first-stage decisions determine the deployment of heterogeneous green vessels on network arcs before demand is realized, and second-stage recourse decisions allocate freight flows through predefined transportation plans after demand scenarios are observed. The objective aims to minimize the total expected cost, consisting of vessel deployment costs and shipping costs, while ensuring full demand satisfaction and compliance with capacity constraints. To evaluate model performance, we design a series of computational experiments that reflect the characteristics of trunk maritime shipping networks. Results demonstrate the tractability of the model for small- to large-scale cases, reveal the trade-offs among different green vessel types. The study contributes to the literature on sustainable shipping optimization and provides practical guidance for the design of cleaner maritime logistics networks under uncertainty.

1. Introduction

Maritime shipping is the backbone of international trade, carrying more than 80 % of the global merchandise volume according to a report by the United Nations Conference on Trade and Development (UNCTAD, 2022). However, despite its critical role in connecting supply chains across continents, the shipping industry has long been recognized as a major contributor to greenhouse gas emissions and air pollutants. The International Maritime Organization (IMO, 2020) reports that international shipping accounts for nearly 3 % of global CO₂ emissions, a share comparable to that of some large, industrialized countries. With the rapid growth of global trade and the rising urgency of climate change mitigation, reducing emissions from maritime transport has become an indispensable priority for policymakers and industry stakeholders (Wu et al., 2023, Yuan et al., 2024). In response, the IMO has introduced increasingly stringent regulations, such as the 2020 sulfur cap and the upcoming carbon intensity indicator (CII) requirements. These

regulatory measures compel ship operators to adopt cleaner and more energy-efficient technologies. Among the available options, three types of green vessels stand out: scrubber-equipped vessels that remove sulfur oxides from exhaust gases; dual-fuel retrofitted vessels capable of switching between conventional marine fuel and liquefied natural gas (LNG); and biofuel-powered single-fuel vessels that rely on renewable energy sources. Each of these vessel categories differs significantly in terms of investment cost, operational expenses, carrying capacity, and emission reduction potential. As a result, fleet planning in modern maritime logistics must balance not only cost efficiency and service reliability but also environmental performance under regulatory and market pressures.

At the same time, maritime shipping demand is inherently uncertain. Global supply chains are increasingly exposed to volatility in trade flows, unexpected demand surges, and disruptions caused by geopolitical tensions, pandemics, or extreme weather events (Wang et al., 2014, Zhao et al., 2021, Wang et al., 2025). Demand uncertainty poses

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^{*} Corresponding author.

E-mail addresses: wangxiuwen3w@163.com (X. Wang), lzhen_shu@163.com (L. Zhen).

significant challenges to fleet deployment: if too few vessels are deployed, demand shortfalls cannot be satisfied; if too many vessels are committed, fixed costs increase unnecessarily, and the environmental footprint may also rise (Yang and Zhen, 2025). Therefore, robust and adaptive decision-making approaches are required to handle the interplay between green vessel technologies and uncertain demand conditions. The academic literature has extensively studied fleet deployment, vessel scheduling, and liner shipping optimization. Traditional research has primarily focused on minimizing cost, maximizing service level, or integrating schedule design with container routing. More recently, attention has shifted toward environmental objectives, with studies examining emission control areas, speed optimization for fuel savings, and carbon pricing policies. However, relatively few works explicitly incorporate the heterogeneity of green vessels into fleet deployment decisions, particularly in the context of uncertain demand. Existing models often assume a homogeneous fleet or ignore the trade-offs among different green technologies. This creates a gap in both theory and practice: how should ship operators deploy a heterogeneous fleet of green vessels across a shipping network under stochastic demand scenarios.

To bridge this gap, this paper develops a two-stage stochastic programming model for cleaner maritime logistics. In the first stage, the operator decides on the deployment of different vessel types across arcs, where each arc is defined not only by the origin–destination (OD) port pair but also by the vessel type serving it. In the second stage, once demand uncertainty is realized, freight flows are allocated through predefined transportation plans consisting of multiple arcs. The objective is to minimize the total expected cost, which combines fixed vessel deployment costs and variable shipping costs, while ensuring demand satisfaction and compliance with capacity constraints. This modeling framework explicitly captures the heterogeneity of green vessels and their economic–environmental trade-offs, as well as the structural uncertainty of demand. To evaluate the proposed model, we design a series of computational experiments on benchmark instances that mimic the characteristics of trunk maritime shipping networks. Six groups of instances are generated, covering small-, medium-, and large-scale cases with varying numbers of OD pairs, arcs, transportation plans, and demand scenarios. Results demonstrate that the model is tractable for small-scale cases and can be solved within reasonable computational time for larger instances using commercial solvers. Furthermore, the experiments reveal how the optimal vessel mix changes under different demand conditions, highlighting the role of biofuel vessels versus dual-fuel and scrubber-equipped vessels. These insights provide practical guidance for ship operators to balance cost efficiency with sustainability goals.

The remainder of this paper is organized as follows. Section 2 reviews the related literatures. Section 3 presents the problem description and formulates the two-stage stochastic programming model. Section 4 details the experimental design. Section 5 reports the computational results and provides managerial insights regarding the deployment of heterogeneous green vessels under demand uncertainty. Finally, Section 6 concludes the study and outlines future research directions.

2. Literature review

In this section, we review relevant literature from two perspectives. The first focuses on green technology adoption and evaluation in maritime shipping, highlighting how emission-control technologies and alternative fuels are assessed and incorporated into shipping operations. The second addresses fleet deployment and scheduling, which examines how heterogeneous fleets are planned and allocated across shipping networks. Together, these two research streams lay the foundation for our study and reveal a key research gap: the lack of network-level, quantitative decision-making models that integrate heterogeneous green vessel technologies with stochastic demand.

2.1. Green technology adoption and evaluation in maritime shipping

A rich body of literature evaluates and optimizes emission-control technologies and alternative fuels in maritime transport. Yang et al. (2012) developed a subjective universal method as a assessment tool for shipowners to evaluate and determine suitable emission-control technologies. Along similar lines, Brynolf et al. (2014) assessed the compliance potential of SO_x and NO_x emissions for three alternative abatement options in future emission control areas (ECAs), namely marine gas oil, scrubbers, and LNG. Sys et al. (2016) analyzed the potential effects of emerging international maritime emission regulations on seaport competition and the economic motivations for establishing emission control areas (ECAs). They concluded that strict sulfur emission limits are influenced more by petrochemical lobbying than by environmental advocacy, and that port shifts from Northern to Southern Europe are unlikely. Complementing these observations, Atari and Prause (2017) investigated the evaluation of scrubbers and identified optimal investment opportunities and the highest returns between fuel switching and scrubber installation. Zhen et al. (2018) proposed an MIP model that helps reduce fuel costs under ECA regulations, highlighting the growing importance of technology choices for regulatory compliance. Beyond technical considerations, Metzger and Schinas (2019) introduced a fuzzy real options approach to appraise green shipping investments, showing that financing schemes such as shared savings can substantially influence the net present value and risk profile of adoption decisions. The exploration of alternative fuels and routes adds further depth. Atilhan et al. (2021) provided a comprehensive review of green hydrogen as a potential maritime fuel, analyzing production pathways, techno-economic performance, storage, and safety issues. Their findings underscore both the environmental advantages of hydrogen and the significant economic and technical barriers to large-scale adoption. Several studies have also focused on the application of green technologies at the route or fleet level. Theocharis et al. (2019) developed an optimization model to evaluate the feasibility of the Northern Sea Route (NSR), aiming to minimize the required freight rate for oil product tanker transport. Their analysis considered forthcoming environmental policies such as low-sulphur fuel requirements and possible heavy-fuel bans, highlighting key economic and regulatory factors influencing NSR adoption. Tseng et al. (2021) developed a quantitative assessment model to analyze the optimal shipping allocation between the Northeast Passage and the traditional Asia–Europe Suez Canal route by considering both operating costs and the external costs of pollutant emissions. Their sensitivity analysis shows that factors such as the icebreaker fee and Suez Canal toll have a stronger influence than fuel price on carriers' adoption decisions for the Northeast Passage. At a higher level of decision-making, Patricksson et al. (2015) also addressed the fleet renewal problem under ECA requirements, although their analysis focused exclusively on scrubbers. Ölçer and Ballini (2015), in turn, developed a broader decision framework that incorporates multiple criteria for evaluating alternative green technologies at the port level. Zhang et al. (2024) investigated the joint decision of green technology adoption and sailing patterns for coastal container ships under ECA regulations, revealing that ignoring endogenous sailing speed and routing may bias investment choices. Industry structure and policy environment have also become focal points. Shang et al. (2024) analyzed the impact of shipping alliances on green technology investments and found that alliances can promote adoption through economies of scale and spill-over effects, though overall welfare depends on the balance between costs and emission reductions. From a policy perspective, Song et al. (2024) employed social network analysis to trace the evolution of China's green shipping policies, demonstrating how policy networks facilitate the diffusion of green practices across stakeholders. Finally, methodological innovations have also been explored. Huang et al. (2022) reviewed the applications of machine learning in sustainable ship design and operation, highlighting its potential to improve energy efficiency, voyage planning, and operational

performance, and arguing that ML can complement traditional optimization approaches in supporting green technology adoption.

In summary, while these studies provide valuable insights into technology assessment, alternative fuels, policy drivers, and methodological advances, most focus on single technologies, individual routes, or specific policy contexts. Few consider the combined effects of multiple green vessel types on network-wide operations or incorporate stochastic demand into the decision-making process. In particular, the heterogeneity of green vessels in terms of cost, capacity, and environmental performance is rarely integrated with scenario-based freight allocation in a unified optimization model. Addressing this gap, our study develops a two-stage stochastic programming framework that jointly optimizes heterogeneous green-vessel deployment and freight flows at the network level under demand uncertainty.

2.2. Fleet deployment and scheduling

Another important research stream focuses on fleet deployment and scheduling. This line of work is often coupled with speed and routing optimization and is increasingly oriented toward environmental goals. Foundational work by Wang and Meng (2017) and Christiansen et al. (2020) provides extensive overviews of maritime fleet planning. Building on this, Wang et al. (2018) formulated the shipping chartering decision as an operational fleet configuration problem integrating fleet deployment and speed optimization under market uncertainty. Baykassoglu et al. (2021) reviewed fleet-planning research and proposed an intermodal planning framework that considers decision levels, multiple stakeholders, and sustainability objectives. With growing environmental pressure, attention has shifted to energy efficiency and emission reduction. Zhen et al. (2020) and Wang et al. (2021) enriched models by introducing various measures aimed at reducing emissions. Zhen et al. (2020) investigated the use of scrubbers and shore power technologies, while Wang et al. (2021) integrated fleet deployment, routing, speed optimization, and schedule design with particular emphasis on the impacts of ECAs. Chen et al. (2022) developed a robust optimization model for the co-deployment of liner alliance fleets operating within a vessel-pool framework under uncertain container transport demand. Through a six-route case study of three liner companies, they showed that joint deployment maximizes alliance profits and effectively leverages shared fleet resources to enhance competitiveness. Research has also expanded toward heterogeneous fleet management. Wang and Wang (2021) examined the joint deployment of heterogeneous fleets across multiple routes. Their study optimized ship selection, order scheduling and sequencing, timetable design, and speed decisions with the aim of minimizing cost. Ribeiro et al. (2021) developed an approximate dynamic programming model to determine optimal heterogeneous fleet sizes for offshore supply vessels, explicitly accounting for uncertain sea conditions and future demand. Hellsten et al. (2022) addressed heterogeneous feeder-vessel network design with simultaneous pickups and deliveries, weekly departures, and demand rejection at a penalty cost. They proposed a branch-and-price framework that solves real instances to optimality, demonstrating its effectiveness for heterogeneous fleet sizing and routing in regional liner shipping. Chen et al. (2023) studied a maritime rescue covering-time minimization problem in Hong Kong waters, deploying a heterogeneous fleet of marine police vessels to minimize emergency response times. They formulated a mathematic model and developed a two-stage algorithm, demonstrating through case studies that their approach efficiently supports real-world heterogeneous fleet rescue planning. Zhao et al. (2024) further extended this stream by proposing a two-step approach for heterogeneous vessel deployment and reliable scheduling under uncertainty, integrating Conditional Value-at-Risk to mitigate navigation and port delay risks. Beyond cost minimization, scholars have begun to adopt multi-objective frameworks that explicitly account for environmental goals. Du et al. (2024) developed a mathematical optimization model for heterogeneous fleet deployment and scheduling that incorporates four green

technologies. Using the non-dominated sorting genetic algorithm II, they derived Pareto-optimal solutions balancing different indicators and demonstrated through a real case study that ship size and age significantly influence technology adoption choices. Similarly, Xiang et al. (2024) studied liner fleet deployment under demand uncertainty, with a particular focus on container redistribution across routes. They introduced a two-stage robust optimization model and solved it using a column-and-constraint generation approach. Their results show that robust optimization provides stable solutions with reduced computational time compared to stochastic programming, making it a practical tool for large-scale applications. In addition, financing mechanisms have also been recognized as critical in shaping fleet deployment decisions. Schinas and Sonechko (2022) proposed a pay-as-you-use business model that integrates operational leasing with outcome-based contracts, thereby alleviating the high upfront investment barrier of green technology adoption. Their results emphasize that financing schemes can significantly affect the feasibility and risk-sharing of green fleet deployment.

In summary, this research stream has evolved from traditional cost-minimization to integrated frameworks that incorporate emissions, uncertainty, and heterogeneity, and even innovative financing mechanisms. Nevertheless, most studies still treat green technology adoption as an external constraint rather than an endogenous part of fleet planning. Few works jointly optimize heterogeneous green vessel deployment and scenario-based freight allocation within a unified two-stage framework. Addressing this gap, our study develops a stochastic programming model that captures both economic and environmental objectives in a network-wide context.

3. Problem description

We consider a green shipping network in which a platform operator is responsible for deploying vessels and allocating freight flows to satisfy uncertain demand. The network is indicated by the set of arcs, where each arc corresponds to a directed service operated by a specific vessel type between two ports. Thus, even if two arcs connect the same pair of ports, they are treated as distinct if they are served by different vessel types, reflecting the heterogeneity in capacity, cost, and environmental performance. Three categories of green vessels are considered: scrubber-equipped vessels, dual-fuel retrofitted vessels, and biofuel single-fuel vessels. Each type is characterized by a fixed deployment cost per vessel, a maximum carrying capacity measured in twenty-foot equivalent units (TEUs), and distinct implications for fuel consumption and emissions. To fulfill demand between origin–destination (OD) pairs, a set of transportation plans is predefined, with each plan corresponding to a feasible route consisting of a sequence of arcs, possibly involving transshipments. The realized demand for each OD pair is uncertain and is modeled by a finite set of scenarios with associated probabilities. The operator faces two interdependent decisions. In the first stage, vessels of different types must be deployed on arcs before demand is realized, which determines the available network capacity. In the second stage, after scenario demand is observed, freight flows are assigned to the predefined plans. Variable shipping costs are incurred in proportion to the amount of cargo transported via each plan, while fixed deployment costs and vessel type choices capture the trade-offs between economic and environmental considerations. The objective aims to minimize the total expected cost, consisting of vessel deployment costs and expected shipping costs, while ensuring that all demand is met in every scenario and that flows do not exceed the capacity provided by the deployed vessels.

The model is based on several practical assumptions. First, it is assumed that the three green vessel types and their technical parameters, such as maximum TEU capacities, weekly deployment costs, and emission characteristics, are known and remain stable throughout the planning horizon. Second, the feasible transportation plans of each OD pair, including the set of constituent arcs and potential transshipments,

are predetermined and fixed, reflecting long-term service agreements and established shipping routes. Third, future demand for each OD pair is expressed by a discrete set of scenarios with known probabilities, that are estimated from historical data and market forecasts and remain constant during the planning horizon. These assumptions ensure that the proposed two-stage stochastic model captures the key economic and environmental trade-offs while remaining computationally tractable.

4. Model

To formalize the problem described above, we propose a two-stage stochastic programming model that captures the deployment of heterogeneous green vessels and the allocation of freight flows under demand uncertainty. The model explicitly incorporates vessel deployment costs, shipping costs, and arc capacity constraints, thereby reflecting the trade-offs between economic efficiency and environmental performance. In the first stage, the decision maker determines the number of different types of vessels deployed on each arc before demand realization. In the second stage, once a demand scenario is observed, freight flows are allocated through predefined transportation plans, incurring variable shipping costs. The objective aims to minimize total expected cost, while guaranteeing that all demands are met and the assigned flows do not exceed the capacity provided on each arc. The mathematical formulation is presented as follows, starting with the notation of sets, parameters, and decision variables.

4.1. Notation

For building the model, the notation for the known parameters and the unknown variables is defined as follows.

Sets and indices

N set of OD pairs, indexed by n .

R_n set of transportation plans of OD pair n , indexed by r .

C set of arcs, indexed by c .

T set of vessel types, indexed by t .

Ω set of scenarios, indexed by ω .

Parameters

f_t fixed cost for deploying a vessel of type t .

q_t maximum capacity of a type t vessel (TEUs).

$d_{n\omega}$ demand of OD pair n under scenario ω .

v_{rc} equal to one if transportation plan r uses arc c and zero otherwise.

p_ω probability of scenario ω .

g_r unit shipping cost via transportation plan r .

Decision variables

α_{tc} integer, number of type t vessels used on arc c .

$\gamma_{r\omega}$ continuous, actual number of TEUs transported via transportation plan r under scenario ω .

4.2. Model formulation

$$[\infty] \text{Min} \sum_{t \in T} \sum_{c \in C} f_t \alpha_{tc} + \sum_{\omega \in \Omega} p_\omega \sum_{n \in N} \sum_{r \in R_n} g_r \gamma_{r\omega} \quad (1)$$

subject to:

$$\sum_{n \in N} \sum_{r \in R_n} \gamma_{r\omega} = d_{n\omega} \quad \forall \omega \in \Omega, n \in N \quad (2)$$

$$\sum_{n \in N} \sum_{r \in R_n} v_{rc} \gamma_{r\omega} \leq \sum_{t \in T} q_t \alpha_{tc} \quad \forall \omega \in \Omega, c \in C \quad (3)$$

$$\alpha_{tc} \geq 0 \quad \forall t \in T, c \in C \quad (4)$$

$$\gamma_{r\omega} \geq 0 \quad \forall \omega \in \Omega, n \in N, r \in R_n. \quad (5)$$

The objective function (1) minimizes the total expected cost, which consists of two components: the first stage fixed deployment cost of

assigning vessels of different types to arcs, and the second stage expected shipping cost incurred when transporting demand through transportation plans under different scenarios. Constraints (2) ensures all demand are satisfied. Constraints (3) enforces arc capacity limitations by guaranteeing that, in each scenario, the total demand flow routed through an arc does not exceed the aggregate capacity provided by the vessels. Constraints (4) and (5) define the domains of decision variables.

5. Computational experiments

To evaluate the effectiveness of the proposed model, we conduct several computational experiments. All experiments are performed on a workstation equipped with an Intel Xeon E5-2643 v3 CPU @ 3.40 GHz and 128 GB of RAM. The algorithms are implemented in Python (version 3.11.5), and all optimization models are solved using GUROBI 12.0.

5.1. Experimental setting

In the experimental study, parameter settings are designed to reflect the characteristics of typical trunk maritime shipping networks, so as to evaluate the performance of the proposed model. A set of major ports is selected to generate bi-directional OD pairs, thus forming a multi-origin and multi-destination transport network, with several feasible transportation plans predefined for each OD pair; the relationship between transportation plans and arcs is represented by binary parameters. On the vessel side, three categories of green ships are considered, namely scrubber-equipped vessels (S), dual-fuel retrofitted vessels (D), and biofuel single-fuel vessels (B). Their capacities and costs are aligned with the ranges reported for Panamax and ultra-large container ships as well as retrofit and alternative-fuel cost studies and are set in the range of 10,000–14,000 TEUs with fixed deployment costs converted to approximately 0.92–1.28 million USD per week, thereby capturing the trade-offs between green technology choices in terms of capacity and cost. Transportation costs for each plan are determined not only by the distances of its constituent arcs and the number of transshipments, but also by the type of vessel operating on those arcs and are restricted to fall within the range of 50–80 USD per TEU, thereby reflecting the combined effects of voyage length, operational complexity, and technological heterogeneity among green vessels. Demands are uniformly distributed within the interval [2000, 9000] TEUs and modeled through scenarios with equal probabilities. To evaluate model performance under different scales, six groups of instances are designed and shown in Table 1.

5.2. Computational performance analysis

To evaluate the computational performance of the proposed model, we test it on a series of benchmark instances of different scales. The experiments are designed to assess the tractability of the two-stage stochastic programming formulation when solved directly by a commercial solver. We report solution time, objective value, and the composition of vessel deployment across different types. By comparing results across instance groups, we are able to understand how problem size and scenario number influence solver efficiency and solution structure. The detailed results of the small-scale instances (ISG1 and ISG2) are reported in Table 2. The model can be solved to optimality within 0.1–0.3 s. The optimal objective values range from 7.36 to 11.74 million USD for ISG1 and from 17.70 to 19.69 million USD for ISG2. On average, about 3 scrubber-equipped vessels, 4 dual-fuel vessels, and 4 biofuel vessels are deployed across these instances. These results demonstrate that the model is highly tractable at small scales and that the fleet mix solutions reflect heterogeneous vessel characteristics. For the medium-scale instances (ISG3 and ISG4), the results are summarized in Table 3. The solution time increases to 1.4–16.6 s, yet the solver can still obtain optimal solutions efficiently. Objective values lie between 30.29 and 32.97 million USD for ISG3, and between 45.37 and 46.87 million USD for ISG4. On average, 7 scrubber-equipped vessels, 13 dual-

Table 1
Scale of instance groups in experiments.

Instance ID	Number of OD pairs	Number of arcs	Number of plans	Number of scenarios	Number of decision variables	Number of constraints
ISG1	5	5	3	5	1×10^1	1×10^1
ISG2	10	8	3	8	1×10^2	1×10^2
ISG3	20	10	5	10	1×10^3	1×10^2
ISG4	30	20	8	20	1×10^3	1×10^3
ISG5	50	30	10	30	1×10^4	1×10^3
ISG6	60	50	12	50	1×10^4	1×10^3

Table 2
Results for small-scale instances (ISG1 and ISG2).

Instance ID	Time (s)	Obj (million USD)	Solution for model		
			Number of type S vessels	Number of type D vessels	Number of type B vessels
ISG1-1	0.1	10.35	2	3	3
ISG1-2	0.1	7.36	2	0	4
ISG1-3	0.1	8.79	1	4	1
ISG1-4	0.1	11.74	3	1	6
ISG1-5	0.1	9.37	4	2	1
ISG2-1	0.2	19.27	4	6	4
ISG2-2	0.2	17.70	3	4	6
ISG2-3	0.3	19.15	3	6	5
ISG2-4	0.2	18.06	1	7	5
ISG2-5	0.2	19.69	5	5	5
Average			3	4	4

Note: “S” indicates scrubber-equipped vessels; “D” indicates dual-fuel retrofitted vessels; “B” indicates biofuel single-fuel vessels.

Table 3
Results for medium-scale instances (ISG3 and ISG4).

Instance ID	Time (s)	Obj (million USD)	Solution for model		
			Number of type S vessels	Number of type D vessels	Number of type B vessels
ISG3-1	2.4	32.24	9	10	3
ISG3-2	4.5	32.97	7	12	4
ISG3-3	1.8	31.15	10	7	5
ISG3-4	1.4	31.64	8	4	11
ISG3-5	1.5	30.29	8	5	9
ISG4-1	9.8	46.06	7	12	13
ISG4-2	16.6	45.37	2	18	10
ISG4-3	14.3	46.87	5	21	4
ISG4-4	15.9	45.74	6	18	6
ISG4-5	14.9	45.50	3	18	10
Average			7	13	8

Note: “S” indicates scrubber-equipped vessels; “D” indicates dual-fuel retrofitted vessels; “B” indicates biofuel single-fuel vessels.

fuel vessels, and 8 biofuel vessels are used, indicating a larger role of dual-fuel vessels as demand and network size expand. The results in Table 4 confirm that the model remains computationally tractable for larger instances. These results also provide meaningful insights into the deployment of heterogeneous green vessels. The detailed results are summarized in Table 3.

The above results demonstrate that optimal fleet deployment patterns are highly sensitive to the distribution of demand scenarios and cost structures. In particular, the share of dual-fuel and biofuel vessels varies markedly as demand shifts, while the total expected cost responds nonlinearly to changes in scenario probabilities. These findings suggest two key managerial implications. First, shipping operators should maintain a diversified green fleet composition to hedge against demand uncertainty, ensuring that capacity remains flexible and compliant with environmental regulations. Second, proactive scenario planning and the incorporation of robust decision rules can significantly reduce both cost risk and emission exposure, thereby supporting long-term sustainability

Table 4
Results for large-scale instances (ISG5 and ISG6).

Instance ID	Time (s)	Obj (million USD)	Solution for model		
			Number of type S vessels	Number of type D vessels	Number of type B vessels
ISG5-1	282.1	72.85	10	26	12
ISG5-2	283.9	75.98	6	40	1
ISG5-3	237.1	73.82	1	43	0
ISG5-4	240.1	71.87	9	29	8
ISG5-5	225.4	74.22	4	40	1
ISG6-1	1145.2	88.80	19	24	17
ISG6-2	1183.5	87.67	4	26	31
ISG6-3	1250.7	89.98	5	23	36
ISG6-4	1785.2	86.58	16	25	18
ISG6-5	1677.8	89.04	15	21	27
Average			9	30	15

Note: “S” indicates scrubber-equipped vessels; “D” indicates dual-fuel retrofitted vessels; “B” indicates biofuel single-fuel vessels.

and competitiveness.

5.3. Managerial insights from fixed cost sensitivity

To examine the impact of fixed cost variations of three vessel types on total cost and the deployment of vessels, we conduct a sensitivity analysis with respect to the weekly fixed costs of the three vessel types. Specifically, the baseline costs of scrubber-equipped (S), dual-fuel (D), and biofuel (B) vessels are increased by 10 %, 20 %, 30 %, and 40 %, respectively, while keeping other parameters unchanged. For each scenario, the model is solved, and the resulting fleet composition and total cost are recorded. This design enables us to isolate the marginal effects of cost variations and to identify substitution patterns among heterogeneous green vessels.

Figs. 1 and 2 illustrate the impact of fixed costs of three vessel types on total cost and fleet deployment. First, when the fixed cost of S vessels increases, the total cost is unchanged. This somewhat counterintuitive result arises because the model entirely abandons S vessels once their cost is perturbed, despite their deployment in the baseline case. In other words, S vessels are marginal in the optimal solution and are easily substituted by other vessel types without increasing total cost. Second, when the fixed costs of D or B vessels increase, the total cost rises initially but stabilizes once the cost reaches a 20 % increase. This plateau

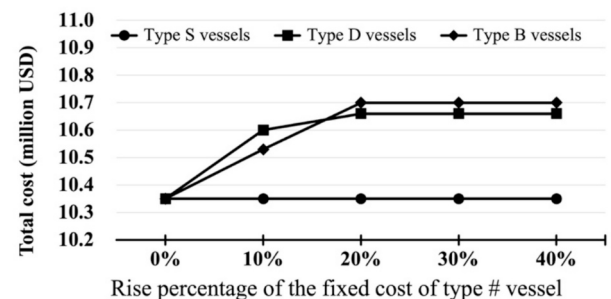


Fig. 1. Impact of fixed cost variations of three vessel types on total cost.

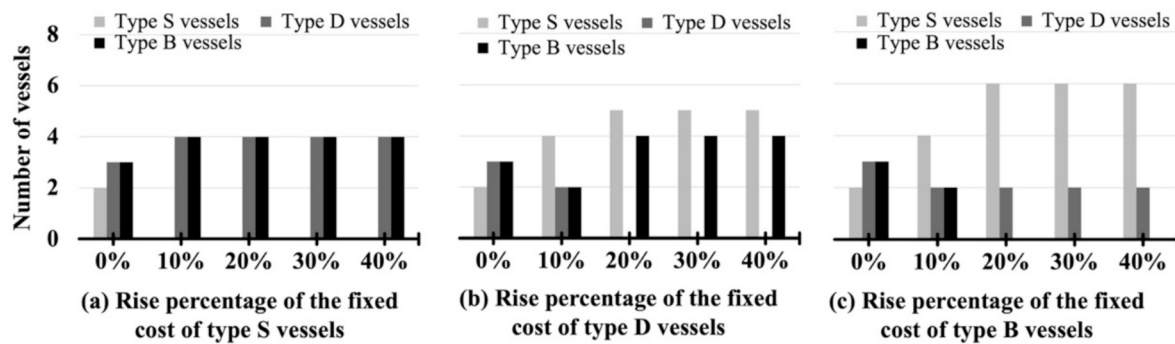


Fig. 2. Impact of fixed cost of three vessel types on the deployment of vessels.

occurs because the model ceases to deploy D or B vessels beyond this threshold, reallocating capacity to the remaining vessel types. The substitution, however, comes at a higher baseline cost, which explains the observed increase in the objective value relative to the baseline.

6. Conclusion

This study investigates fleet deployment and freight allocation for cleaner maritime logistics under demand uncertainty. A two-stage stochastic programming model is developed, where the first-stage decision determines the deployment of heterogeneous green vessels (i.e., scrubber-equipped, dual-fuel retrofitted, and biofuel single-fuel ships) across network arcs, and the second-stage decision allocates container flows through predefined transportation plans after demand realization. The objective is to minimize the total expected cost, including fixed deployment and variable shipping costs, subject to demand fulfillment and capacity constraints.

From the perspective of modeling, this paper integrates heterogeneous vessel technologies and uncertain market demand into a unified optimization framework. The model highlights the trade-offs between cost efficiency and environmental performance and provides a systematic way to analyze deployment strategies of green vessels under uncertain demand scenarios.

From the perspective of computational performance, extensive experiments are conducted on multiple groups of instances with different scales. Results show that the model can be efficiently solved by a commercial solver for small- and medium-scale networks, while larger-scale cases require longer computation time but remain tractable. Moreover, the allocation patterns of different vessel types reflect their comparative advantages: scrubber-equipped vessels are more frequently deployed on cost-sensitive routes, dual-fuel vessels dominate in high-volume demand cases, and biofuel vessels serve as complementary options to enhance sustainability.

Nevertheless, the study has some limitations. The current model assumes a static single-period decision framework and does not incorporate dynamic deployment strategies. In future research, we will further enrich the stochastic framework by incorporating uncertainty from both the demand side and the supply side (Pratson 2023, Qi et al., 2024a), so as to better capture real-world operational risks. We also intend to explore alternative objective functions, for example by considering multiple objectives such as cost efficiency, service reliability, and environmental performance, so that decision makers can examine the problem from different perspectives and identify the most appropriate deployment strategy under varying priorities (Lestari et al., 2024). In addition, to efficiently handle larger-scale networks that may exceed the capability of commercial solvers, we plan to develop customized decomposition or heuristic algorithms tailored to the structure of the proposed model (Zhou et al., 2024, Qi et al., 2024b).

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CRediT authorship contribution statement

Xiuwen Wang: Writing – original draft. **Lu Zhen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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